

Research Summary & Verified References

Verifiable Chain-of-Custody for Digital Evidence Files

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Annotated Literature Summary

Ten research sources were consulted during the literature survey for this project. Each is summarized below with its core contribution and its relevance to the Verifiable Chain-of-Custody project.

1. B-CoC: A Blockchain-based Chain of Custody for Evidences Management in Digital Forensics (2018)

Authors: Silvia Bonomi, Marco Casini, & Claudio Ciccotelli

Venue: arXiv preprint (Sapienza Università di Roma)

Summary: Proposes a Blockchain-based Chain of Custody (B-CoC) built on a private Ethereum network to dematerialize the traditionally manual, paper-based custody process. Every transfer of evidence between parties is recorded as a blockchain transaction, giving auditable integrity and traceability of ownership. The authors built a working prototype and benchmarked its performance (transaction latency, storage growth) under simulated evidence workloads.

Relevance to this project: Closest prior art in spirit — solves the same tamper-evidence problem — but does so by standing up a private blockchain network, which is exactly the infrastructure overhead this project avoids.

2. Lightweight Tamper-Evident Log Integrity Verification for IoT Edge Environments: A Merkle-Tree Pipeline with Adaptive Chunking (2026)

Authors: Muhammet Anil Yagiz, Fahrettin Horasan, & Ahmet Hasim Yurttakal

Venue: arXiv preprint

Summary: Presents a Merkle-tree-based pipeline with adaptive chunking that achieves tamper-evident log verification for IoT edge devices without any distributed ledger. Reports very high throughput (over 130,000 log entries/second) and small, $O(\log n)$ inclusion proofs, explicitly citing Crosby & Wallach's tamper-evident logging work as its theoretical basis.

Relevance to this project: The closest technical match to this project's approach — proves that Merkle-tree tamper-evidence without blockchain infrastructure is practical at scale. Its gap: built for generic IoT telemetry, not forensic evidence with investigator-attributable signatures.

3. Efficient Data Structures for Tamper-Evident Logging (2009)

Authors: Scott A. Crosby & Dan S. Wallach

Venue: Proceedings of the 18th USENIX Security Symposium

Summary: The foundational paper on Merkle-tree-chained "history trees" for tamper-evident logs, offering $O(\log n)$ proofs of both inclusion and consistency between two versions of a log. This is the theoretical basis nearly every later Merkle-tree tamper-evidence paper (including #2 above) builds on.

Relevance to this project: Core building block for this project's Merkle tree module — the algorithms here directly inform merkle_tree.py's design. Predates any forensic custody or digital-signature application.

[4. Forensic-chain: Blockchain based digital forensics chain of custody with PoC in Hyperledger Composer \(2019\)](#)

Authors: Auqib Hamid Lone & Roohie Naaz Mir

Venue: Digital Investigation

Summary: Proposes "Forensic-chain," a permissioned blockchain framework built on Hyperledger Composer, using hash pointers between blocks to create an append-only, auditable evidence trail. Includes a working proof-of-concept implementation.

Relevance to this project: Reinforces the pattern seen across the field: robust tamper-evidence, but requires standing up and operating a Hyperledger network — not standalone.

[5. Blockchain-based chain of custody: Towards real-time tamper-proof evidence management \(2020\)](#)

Authors: Liza Ahmad, Salam Khanji, Farkhund Iqbal, & Faouzi Kamoun

Venue: Proceedings of the 15th International Conference on Availability, Reliability and Security (ARES '20), ACM

Summary: Proposes a chain-of-custody framework on a private Ethereum blockchain that stores evidence metadata (not the evidence itself) and integrates with physical smart locks, so only authorized parties can access stored evidence. Targets law enforcement, legal, and forensic stakeholders jointly.

Relevance to this project: Shows blockchain CoC extended into the physical world (smart locks) — useful context, but adds blockchain + IoT hardware dependency, increasing deployment complexity beyond what this project needs.

[6. Potential applicability of blockchain technology in the maintenance of chain of custody in forensic casework \(2024\)](#)

Authors: Harsh E. Patil, Ravshish Kaur Kohli, Sorabh Puri, & Pooja Puri

Venue: Egyptian Journal of Forensic Sciences, 14(1)

Summary: A review paper arguing for blockchain's potential to replace manual, logbook-based chain-of-custody records, which the authors describe as time-consuming and prone to tampering. Surveys prior blockchain-forensics work and frames blockchain's immutability and cryptographic guarantees as the key advantage over current practice.

Relevance to this project: Useful for framing the "why it matters" motivation (traditional CoC is manual and fragile), though it stays at the review/conceptual level rather than proposing a concrete lightweight technical design.

[7. Blockchain for Modern Digital Forensics: The Chain-of-Custody as a Distributed Ledger \(2019\)](#)

Authors: Haider M. Al-Khateeb, Gregory Epiphaniou, & Herbert Daly

Venue: Book chapter / conference contribution

Summary: A conceptual piece arguing that evidentiary integrity should be understood as something demonstrated through a verifiable, challengeable record — and that a permissioned distributed ledger is well-suited to producing that record. Frames the chain-of-custody problem in terms courts and legal practitioners would recognize.

Relevance to this project: Provides the legal-admissibility framing this project can draw on when arguing why cryptographic verifiability (not just procedural paperwork) matters — without prescribing blockchain as the only route there.

[8. Blockchain-Based Chain Of Custody: A Secure Digital Evidence Framework For Digital Forensics Investigation \(2022\)](#)

Authors: Febin Prakash, Harsh Sadawarti

Venue: International Journal of Applied Innovative Research (reported), Vol. 3, Issue 1–4

Summary: Proposes a secure digital-evidence framework combining RC4 encryption with fuzzy hash segmentation, referencing Hyperledger Caliper for performance benchmarking and prior work using RC4 to encrypt XML-based custody data.

Relevance to this project: Focuses on encrypting the evidence data itself alongside custody tracking — a different emphasis from this project's focus on tamper-evident event chaining.

[9. Chain of Custody and Evidence Integrity Verification Using Smart Contracts \(2021\)](#)

Authors: Adir Miller, Avinash Singh

Venue: Proceedings of the International Conference on Cyber Warfare and Security (ICCWS)

Summary: Explores using Ethereum smart contracts and wallet-based signing (e.g., MetaMask's `personal_sign`) to record custody events as signed, replay-resistant on-chain transactions, arguing that signed events provide stronger integrity guarantees than hashing alone.

Relevance to this project: Directly supports this project's core thesis that signed events (not just hashes) are needed for strong custody guarantees — but implements that idea via smart contracts and wallet infrastructure rather than a standalone signer.

[10. Blockchain-based Digital Chain of Custody Multimedia Evidence Preservation Framework for Internet-of-Things \(2023\)](#)

Authors: Sakshi, Aruna Malik, Ajay K. Sharma

Venue: Journal of King Saud University – Computer and Information Sciences

Summary: Applies a standard block-header design — Merkle tree, previous-block hash, timestamp — to preserve multimedia evidence captured by IoT devices, targeting domain-specific integrity guarantees for IoT-sourced forensic data.

Relevance to this project: Confirms that the block-header pattern (Merkle tree + hash chain + timestamp) this project uses is an established, recognized structure in forensic literature — just applied here to a narrower IoT domain.